

IC637 Program Analysis

Lecture 7: Context Depth and Flavors

Minseok Jeon

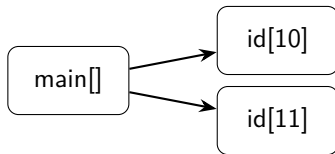
2025 Fall

Review: Necessity of Context Sensitive Analysis

- **1-call-site sensitive** analysis can prove the castings are always safe.
- call-site sensitivity uses the call sites as context elements.

```
1 class A{}
2 class B{}
3 class C{
4     static Object id(Object v) {
5         return v;
6     }
7     void main(String[] args) {
8         A a = new A(); //l1
9         B b = new B(); //l2
10        A a2 = (A)id(a); //query1
11        B b2 = (B)id(b); //query2
12    }
13 }
```

1-call-site sensitive analysis

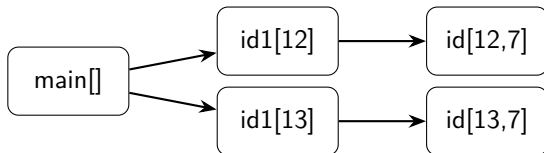

$$\begin{aligned} a[] &\rightarrow \{l_1[]\} & b[] &\rightarrow \{l_2[]\} \\ v[10] &\rightarrow \{l_1[]\} & v[11] &\rightarrow \{l_2[]\} \\ a2[] &\rightarrow \{l_1[]\} & b2[] &\rightarrow \{l_2[]\} \end{aligned}$$

2-Call-Site Sensitive Analysis

- conventional 2-call-site sensitive uses the caller methods' contexts.

```
1 class A{} class B{}
2 class C{
3     static Object id(Object v) {
4         return v;
5     }
6     static Object id1(Object v) {
7         return id(v);
8     }
9     void main(String[] args) {
10         A a = new A(); //l1
11         B b = new B(); //l2
12         A a2 = (A)id1(a); //query1
13         B b2 = (B)id1(b); //query2
14     }
15 }
```

2-call-site sensitive analysis


$$\begin{aligned} a[] &\rightarrow \{l_1[]\} & b[] &\rightarrow \{l_2[]\} \\ id1.v[12] &\rightarrow \{l_1[]\} & id1.v[13] &\rightarrow \{l_2[]\} \\ id.v[12,7] &\rightarrow \{l_1[]\} & id.v[13,7] &\rightarrow \{l_2[]\} \\ a2[] &\rightarrow \{l_1[]\} & b2[] &\rightarrow \{l_2[]\} \end{aligned}$$

Call-Site Sensitive Analysis

- Define **Record** and **Merge** for 1-call-site-sensitivity + 0-context-sensitive heap.

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

- Define **Record** and **Merge** for 1-call-site-sensitivity + 1-context-sensitive heap.

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

- Define **Record** and **Merge** for 2-call-site-sensitivity + 1-context-sensitive heap.

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

- Define **Record** and **Merge** for 2-call-site-sensitivity + 2-context-sensitive heap.

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

Quiz

- How can we prove the queries using conventional 1-context sensitive analysis?

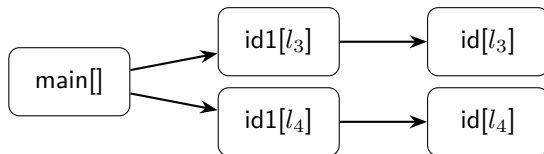
```
1 class A{} class B{}
2 class C{
3     Object id(Object v) {
4         return v;
5     }
6     Object id1(Object v) {
7         return this.id(v);
8     }
9 }
10 class D{
11     void main(String[] args) {
12         A a = new A(); //l1
13         B b = new B(); //l2
14         C c1 = new C(); //l3
15         C c2 = new C(); //l4
16         A a3 = (A)c1.id1(a); //query3
17         B b3 = (B)c2.id1(b); //query4
18     }
19 }
```

Object Sensitivity

- Object sensitivity uses the receiver objects as context elements.

```
1 class A{} class B{}
2 class C{
3   Object id(Object v) {
4     return v;
5   }
6   Object id1(Object v) {
7     return this.id(v);
8   }
9 }
10 class D{
11   void main(String[] args) {
12     A a = new A(); //l1
13     B b = new B(); //l2
14     C c1 = new C(); //l3
15     C c2 = new C(); //l4
16     A a3 = (A)c1.id1(a); //query3
17     B b3 = (B)c2.id1(b); //query4
18   }
19 }
```

1-object sensitive analysis


$$\begin{aligned} a[] &\rightarrow \{l_1[]\} & b[] &\rightarrow \{l_2[]\} \\ v1[l_3] &\rightarrow \{l_1[]\} & v1[l_4] &\rightarrow \{l_2[]\} \\ v2[l_3] &\rightarrow \{l_1[]\} & v2[l_4] &\rightarrow \{l_2[]\} \\ a2[] &\rightarrow \{l_1[]\} & b2[] &\rightarrow \{l_2[]\} \end{aligned}$$

2-Object Sensitivity

- How can we define 2-object sensitivity that can prove the queries?

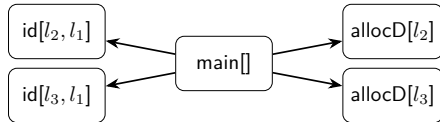
```
1 class A{} class B{}
2 class C{
3     D allocD() {
4         return new D();} //l1
5 }
6 class D{
7     Object id(Object v) {
8         return v;}
9 }
10 class E{
11     void main(String[] args) {
12         C c1 = new C(); //l2
13         C c2 = new C(); //l3
14         D d1 = c1.allocD();
15         D d2 = c2.allocD();
16         A a = (A)d1.id(new A()); //query1
17         B b = (B)d2.id(new B()); //query2
18 }}
```

2-Object Sensitivity

- **2-object sensitive** analysis uses the heap contexts of the receiver objects.

```
1 class A{} class B{}
2 class C{
3     D allocD() {
4         return new D();} //l1
5 }
6 class D{
7     Object id(Object v) {
8         return v;}
9 }
10 class E{
11     void main(String[] args) {
12         C c1 = new C(); //l2
13         C c2 = new C(); //l3
14         D d1 = c1.allocD();
15         D d2 = c2.allocD();
16         A a = (A)d1.id(new A()); //l4
17         B b = (B)d2.id(new B()); //l5
18 }}
```

2-object sensitive analysis


$$\begin{aligned} c1[] &\rightarrow \{l2[]\} & c2[] &\rightarrow \{l3[]\} \\ d1[] &\rightarrow \{l1[l2]\} & d2[] &\rightarrow \{l1[l3]\} \\ v[l2, l1] &\rightarrow \{l4[]\} & v[l3, l1] &\rightarrow \{l5[]\} \\ a[] &\rightarrow \{l4[]\} & b[] &\rightarrow \{l5[]\} \end{aligned}$$

2-Object Sensitivity

- Typical code pattern that needs 2-object sensitivity

```
class A{}
class B{}
class C {
    public static void main (){
        ArrayList al1 = new ArrayList();//AL1
        ArrayList al2 = new ArrayList();//AL2

        al1.add(new A());
        al2.add(new B());

        ArrayList.ListItr it1 = al1.iterator();
        ArrayList.ListItr it2 = al2.iterator();

        A a = (A)it1.next(); //Query 1
        B b = (B)it2.next(); //Query 2
    }
}
```

```
class ArrayList{
    Object[] elementData = new Object[10];
    int size = 0;
    void add(Object e){
        elementData[size++] = e;
    }
    ListItr iterator(){
        return new ListItr(); //IT
    }
    class ListItr{
        int cursor = 0;
        Object next(){
            return elementData[cursor++];
        }
    }
}
```

Object Sensitivity

- Define **Record** and **Merge** for 1-object-sensitivity + 0-context-sensitive heap.

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

- Define **Record** and **Merge** for 1-object-sensitivity + 1-context-sensitive heap.

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

- Define **Record** and **Merge** for 2-object-sensitivity + 1-context-sensitive heap.

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

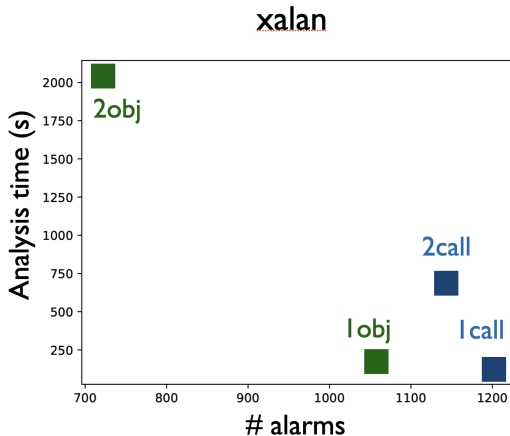
- Define **Record** and **Merge** for 2-object-sensitivity + 2-context-sensitive heap.

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

2-Object Sensitivity

- 2-object sensitivity is known to be highly precise but also very expensive analysis.



Quiz

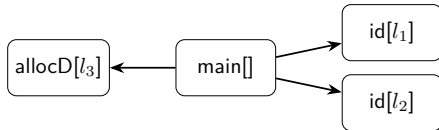
- (2-)object sensitivity is precise but very expensive.
- Try to define a new context flavor which is a faster (but less precise) version of object sensitivity.

Problem of Object Sensitivity

- Problem of object sensitivity in terms of scalability.

```
1 class C{
2   D allocD() {
3     if (cond()){
4       return new D();//l1
5     }
6     else{
7       return new D();//l2
8     }
9   }
10 }
11 class D{
12   Object id(Object v) {
13     return v;}
14 }
15 class E{
16   void main(String[] args) {
17     C c = new C();//l3
18     D d = c.allocD();
19     A a = (A)d.id(new A());//l4
20   }
21 }
```

1-object sensitive analysis

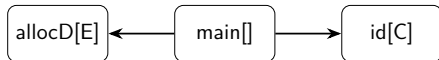

$$\begin{aligned}c[] &\rightarrow \{l_3[]\} \\ d[] &\rightarrow \{l_1[], l_2[]\} \\ v[l_1] &\rightarrow \{l_4[]\} \\ v[l_2] &\rightarrow \{l_4[]\}\end{aligned}$$

Type Sensitivity

- Type sensitivity¹ is a coarser version of object sensitivity.
- Type sensitivity uses class types instead of objects.

```
1 class C{
2   D allocD() {
3     if (cond()){
4       return new D();//l1
5     }
6     else{
7       return new D();//l2
8     }
9   }
10 }
11 class D{
12   Object id(Object v) {
13     return v;}
14 }
15 class E{
16   void main(String[] args) {
17     C c = new C();//l3
18     D d = c.allocD();
19     A a = (A)d.id(new A());//l4
20   }
21 }
```

1-object sensitive analysis


$$\begin{aligned}c[] &\rightarrow \{l_3[]\} \\ d[] &\rightarrow \{l_1[], l_2[]\} \\ v[C] &\rightarrow \{l_4[]\}\end{aligned}$$

¹<https://dl.acm.org/doi/10.1145/1925844.1926390>

Type Sensitivity

- Define **Record** and **Merge** for 1-type-sensitivity + 0-context-sensitive heap where $toClass: \mathbb{H} \rightarrow \mathbb{C}$ maps each heap allocation site to its allocating class (e.g., $toClass(l_1) = C$).

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

- Define **Record** and **Merge** for 1-type-sensitivity + 1-context-sensitive heap.

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

- Define **Record** and **Merge** for 2-type-sensitivity + 1-context-sensitive heap.

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

- Define **Record** and **Merge** for 2-type-sensitivity + 2-context-sensitive heap.

Record(heap, ctx) =

Merge(heap, hctx, invo, callerCtx) =

Quiz

- Try to define a new context flavor that is a coarser version of call-site sensitivity.

Quiz

- Try to define a new context flavor that is a coarser than object sensitivity but more precise than type sensitivity.

Quiz

- When $i < j$, if i -call-site sensitivity proves a query, does j -call-site sensitivity also prove the same query? If so, why? If not, give a counterexample.

- When $i > j$, if i -call-site sensitivity proves a query, does j -call-site sensitivity also prove the same query? If so, why? If not, give a counterexample.

Quiz

- If i -call-site sensitivity proves a query, does j -object sensitivity also prove the same query? If so, why? If not, give a counterexample.

- If i -object sensitivity proves a query, does j -call-site sensitivity also prove the same query? If so, why? If not, give a counterexample.

Quiz

- If k -type sensitivity proves a query, does k -object sensitivity also prove the same query? If so, why? If not, give a counterexample.

- If k -object sensitivity proves a query, does k -type sensitivity also prove the same query? If so, why? If not, give a counterexample.

Quiz

- When $i < j$, if i -object sensitivity proves a query, does j -type sensitivity also prove the same query? If so, why? If not, give a counterexample.